COMP 262: Natural Language Processing and Recommender Systems

Sentiment Analysis of Amazon All Beauty Reviews

Phase 1: Lexicon-based Models (VADER and TextBlob)

Team 2

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1. Introduction

- **Goal:** label Amazon All Beauty reviews as Positive, Neutral or Negative using two lexicon models.
- **Data:** Amazon Review Data (2018), All Beauty 5-core, published by Ni, Li and McAuley (UCSD).
- **Code:** team2_phase1_lexicon_sentiment.py runs every step and saves all charts and tables (seed 42).

2. Data Exploration

Table 1. *Dataset overview*

Rows	Columns	Products	Reviewers	Avg rating	5-star	Missing text
5,269	12	85	991	4.77	88.2%	5 rows

- **Duplicates:** 3,890 rows (73.8%) are the same review copied under different product sizes/colours. Only **1,379 unique reviews** remain.
- **Imbalance:** 91% of unique reviews are positive, so we use **macro-F1**, not accuracy.
- **Products:** median 4 reviews per product; 2 products hold 62.5% of all reviews.
- **Users:** 86% of reviewers wrote only one unique review.
- **Length:** median 29 words; 106 long outliers (> 161 words). 2-3 star reviews are the longest.

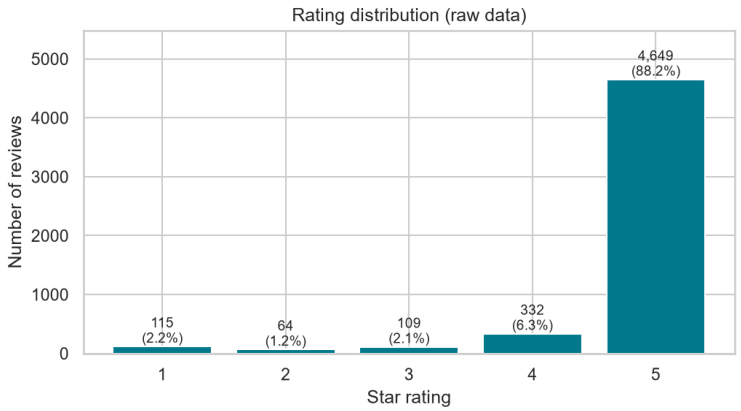


Figure 1. *Rating distribution*

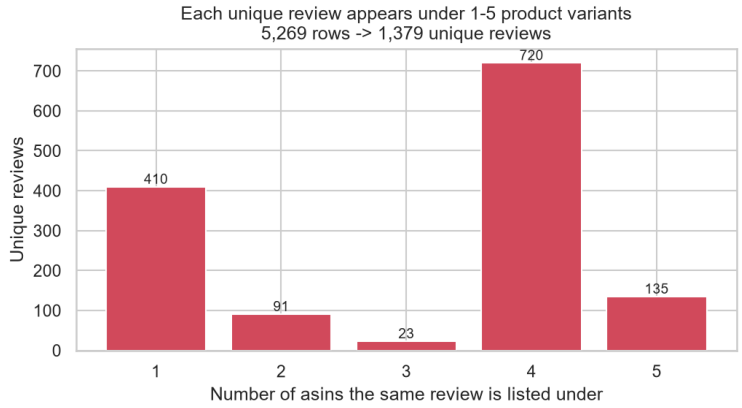


Figure 2. *Product variants each review is copied to*

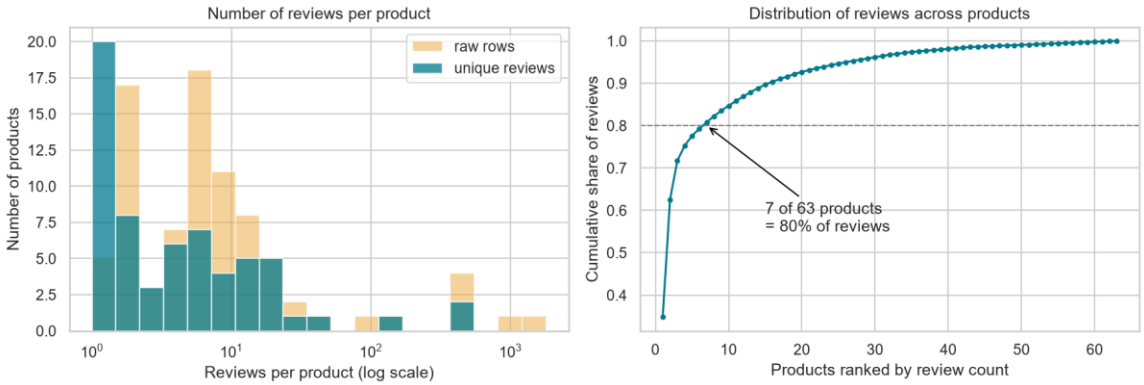


Figure 3. *Reviews per product and their cumulative share*

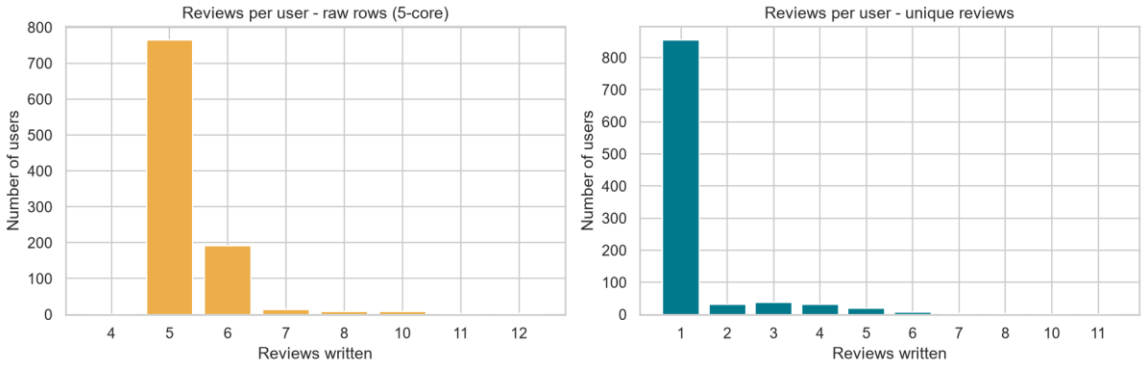


Figure 4. *Reviews per user: raw vs unique*

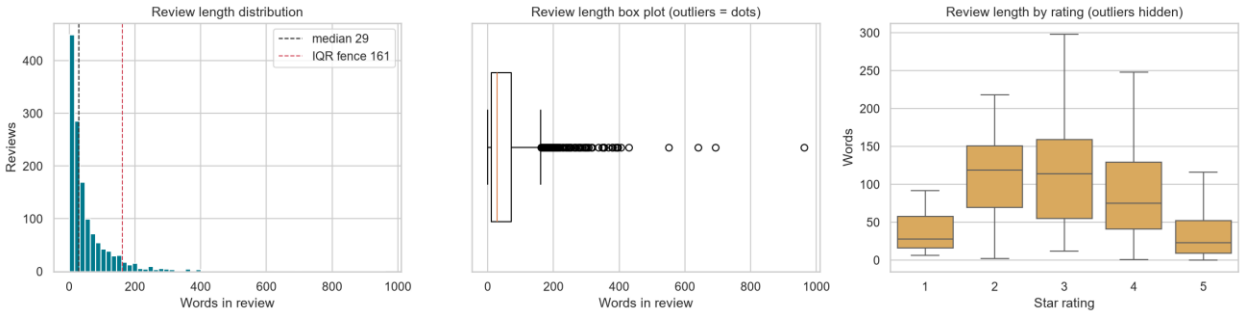


Figure 5. *Review length, outliers and length by rating*

3. Pre-processing

Table 2. *Labels from star ratings*

Label	Ratings	Reviews	Share
Positive	4, 5	1,259	91.3%
Neutral	3	64	4.6%
Negative	1, 2	56	4.1%

- **Columns used:** reviewText + summary as input; overall as the label only. Other columns were used only for exploration.
- **Removed:** duplicate copies, 1 empty review, and auto-generated "Five Stars" titles (they repeat the rating, so they leak the label).
- **Kept:** short and long reviews. They are real opinions, and long ones are mostly the minority classes.

Table 3. *Pre-processing per model*

Step	VADER	TextBlob	Reason
Clean HTML, URLs, "sooooo"	Yes	Yes	Noise
Lower-case, remove punctuation	No	Yes	VADER uses CAPS and "!"
Remove stop words	No	Yes, keep "not"	Negations carry sentiment
Lemmatise	No	Yes	Match TextBlob's base words

- Tested on 378 held-out reviews: this setup gave TextBlob macro-F1 0.506 vs 0.460 with light cleaning. Removing "not" hurt both models.

4. Models

- **Lexicons studied:** VADER, TextBlob, SentiWordNet. We chose **VADER** (handles negation, CAPS, "!!!", emoticons, "but") and **TextBlob** (adjective lexicon with subjectivity). **SentiWordNet** was rejected: it scores "not good" as positive.

Table 4. *The two models*

	VADER	TextBlob
How it works	Rated word list + grammar rules; compound score -1 to +1	Averages adjective scores; flips on negation; polarity -1 to +1
Rule	Pos >= 0.05, Neg <= -0.05	Pos > 0, Neg < 0
Tuned threshold	+/- 0.40	Pos >= 0.075, Neg <= 0.05

- **Test data:** 1,000 random unique reviews (911 Positive, 51 Neutral, 38 Negative). Thresholds were tuned only on the other 378 reviews.

5. Results

Table 5. *Results on the 1,000-review test sample*

Model	Accuracy	Macro-F1	F1 Neg	F1 Neu	F1 Pos
Majority baseline (always Positive)	0.911	0.318	0.000	0.000	0.953
VADER (default thresholds)	0.891	0.469	0.432	0.023	0.951
TextBlob (default thresholds)	0.866	0.404	0.234	0.044	0.933
VADER (tuned (0.4, -0.4))	0.859	0.490	0.361	0.177	0.932
TextBlob (tuned (0.075, 0.05))	0.834	0.391	0.225	0.030	0.919

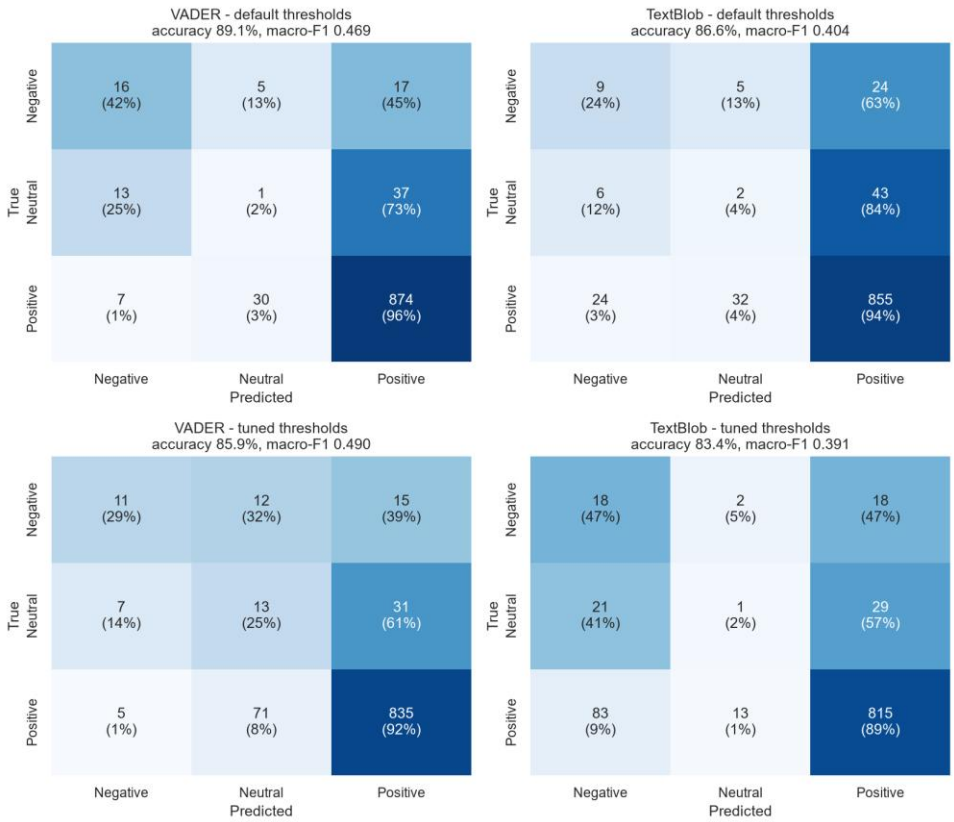


Figure 6. *Confusion matrices (default and tuned thresholds)*

- **VADER wins:** macro-F1 0.469 vs 0.404, and it finds far more negative reviews.
- **Neutral is the weak spot:** 73-84% of neutral reviews are predicted Positive.
- **Long reviews are harder:** accuracy falls from 95% (11-30 words) to 74% (150+ words).
- **Tuning** raised VADER to 0.490 but did not help TextBlob.

6. Conclusion

- De-duplication was essential: 74% of rows were copies.
- VADER is the better lexicon for these reviews.

- Both lexicons miss neutral and mixed reviews, which Phase 2's machine-learning models will target on the same 1,000 reviews.

7. Assumptions

- The star rating is the true sentiment.
- Same reviewer + text + timestamp = the same review.

8. References

[1] Ni, J., Li, J., & McAuley, J. (2019). Justifying recommendations using distantly-labeled reviews and fine-grained aspects. EMNLP-IJCNLP 2019. https://nijianmo.github.io/amazon/index.html

[2] Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. ICWSM-14.

[3] Loria, S. TextBlob documentation. https://textblob.readthedocs.io

[4] Baccianella, S., Esuli, A., & Sebastiani, F. (2010). SentiWordNet 3.0. LREC 2010.

[5] Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python. O'Reilly.

Appendix 1: Project Plan

Table 6. *Project plan*

Week	Task	Owner	Status
3	Teams formed, dataset assigned	All	Done
4-5	Exploration and pre-processing	Manav Chaudhary, Rishi Goyal	Done
6	Lexicon study and per-model cleaning	Steven Hao Gong, Harsh Eary	Done
7	Models and validation	Parth Kishorbhai Kevadiya, Jasdish Singh	Done
8	Report and Phase 1 demo	All	In progress
9-14	Phase 2	All	Planned

Appendix 2: Meeting Register

Table 7. *Meeting register*

Date / time	Attendees	Subjects	Assignments
[dd-mm-yyyy hh:mm]	[names]	Kick-off, dataset	Manav, Rishi: exploration and pre-processing
[dd-mm-yyyy hh:mm]	[names]	Exploration findings	Steven, Harsh: lexicons and cleaning
[dd-mm-yyyy hh:mm]	[names]	Lexicons and results	Parth, Jasdish: models and results